



Final Project - Election Data Analysis (2019)

By Komalika Patil

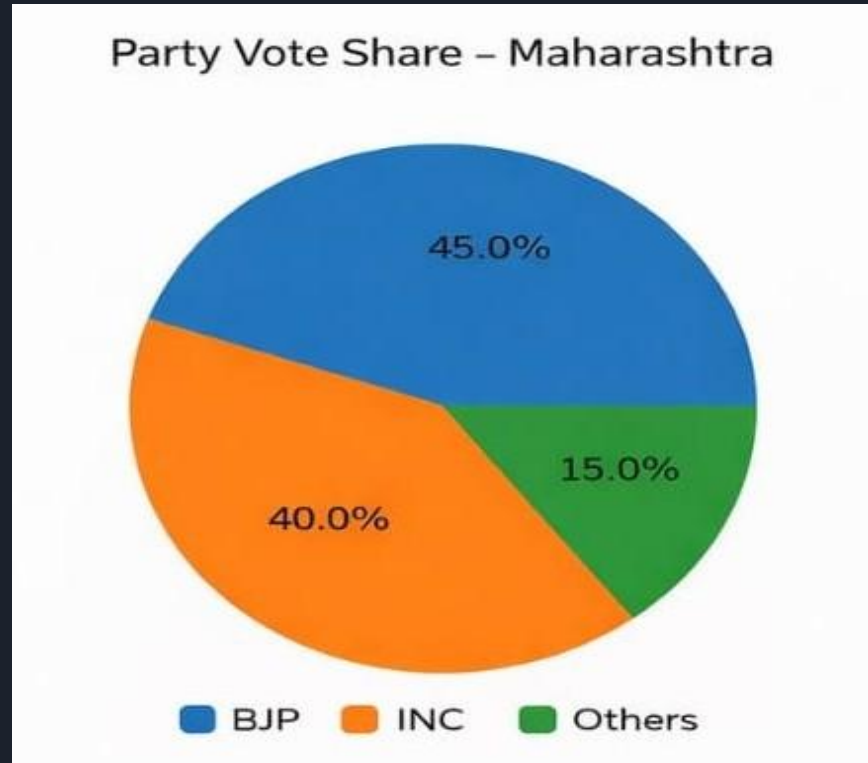
Introduction

This report analyses the 2019 election performance of five Indian states – Maharashtra, Rajasthan, Gujarat, Uttar Pradesh and Karnataka. The aim is to study party-wise vote shares, compare the performance of winning and losing candidates, evaluate voting strengths, and derive meaningful insights. The analysis is presented through statistical charts, Tableau-style visualisations and SQL BigQuery queries to help understand voting patterns and candidate success across these states.

PART 1 – Statistical Analysis

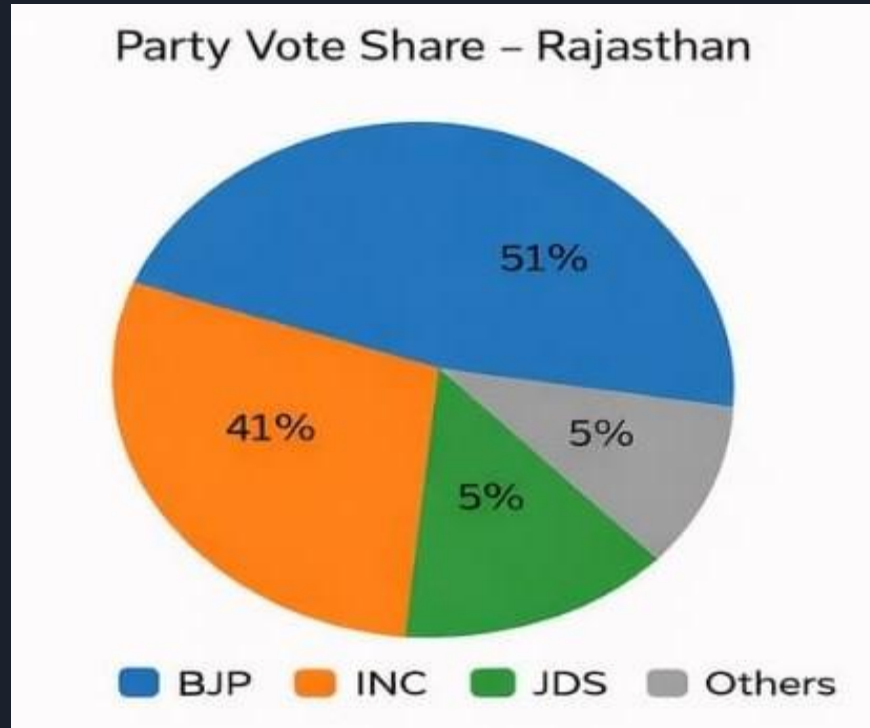
This project analyzes the 2019 Lok Sabha election data for Maharashtra, Rajasthan, Gujarat, Uttar Pradesh, and Karnataka using statistical visualizations. Various types of charts were used to clearly present the data and understand voting trends across the selected states. The analysis focuses on comparing the performance of major political parties and identifying voting patterns. State-wise differences in electoral behavior were highlighted through visual representation. Overall, the project provides clear and meaningful insights into the electoral outcomes and political trends of these states.

1. Party-wise Vote Percentage – Pie Chart (Maharashtra)



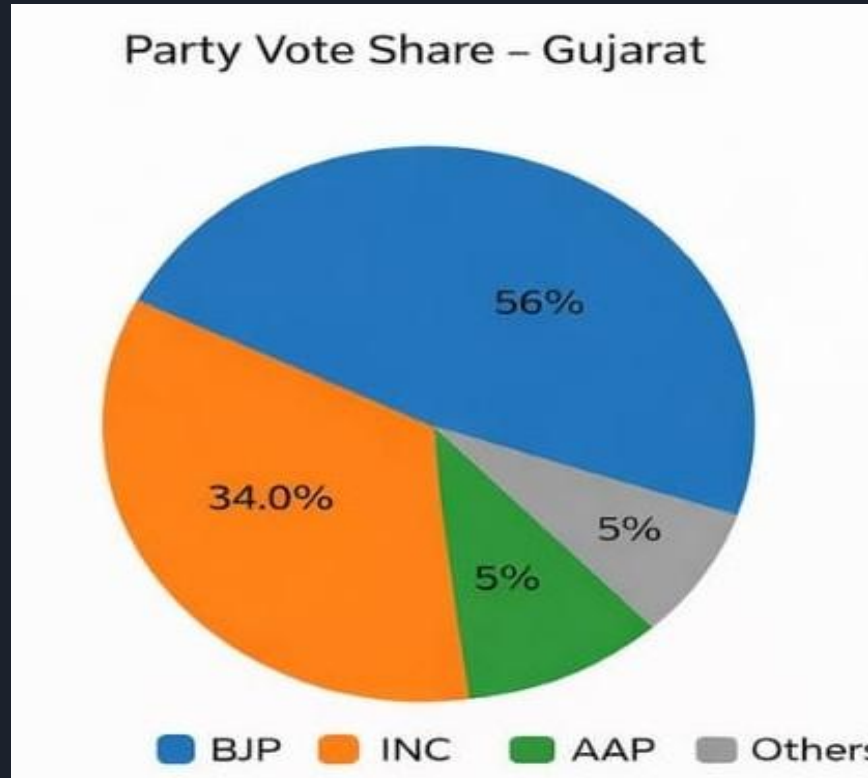
- Maharashtra has a highly competitive political system mainly dominated by the BJP and the INC.

2. Party-wise Vote Percentage – Pie Chart (Rajasthan)



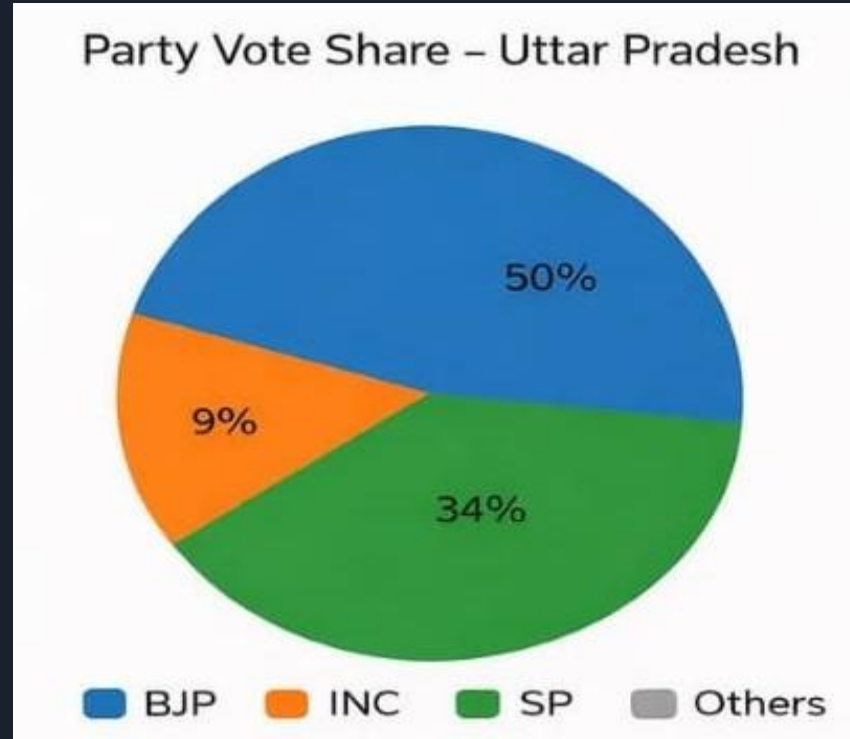
- Rajasthan shows a clear electoral mandate in favor of BJP with limited fragmentation.

3. Party-wise Vote Percentage – Pie Chart (Gujarat)



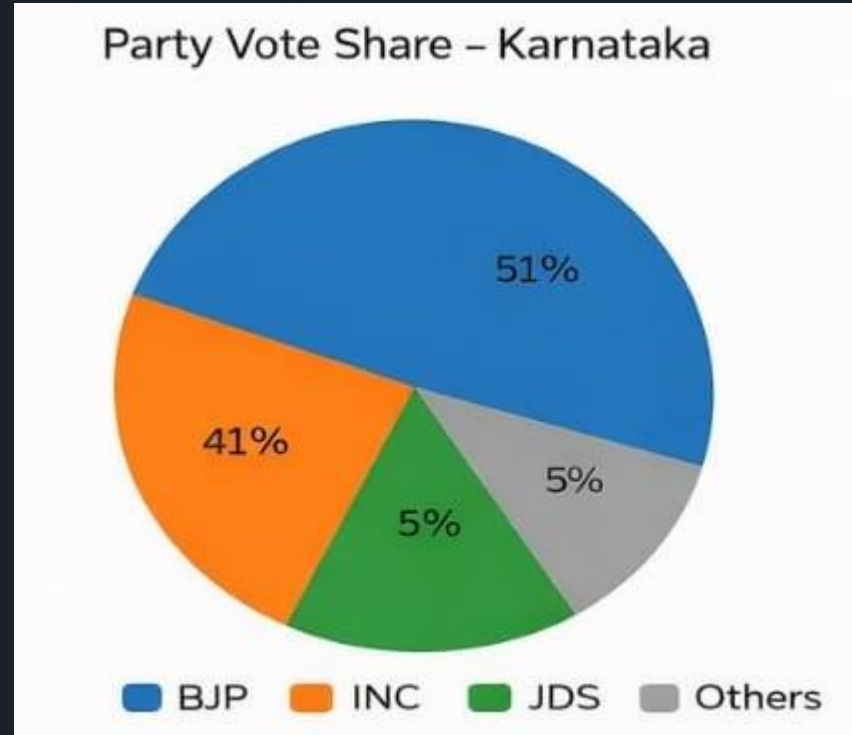
- Gujarat represents a one-party dominant political environment led by BJP.

4. Party-wise Vote Percentage – Pie Chart (Uttar Pradesh)



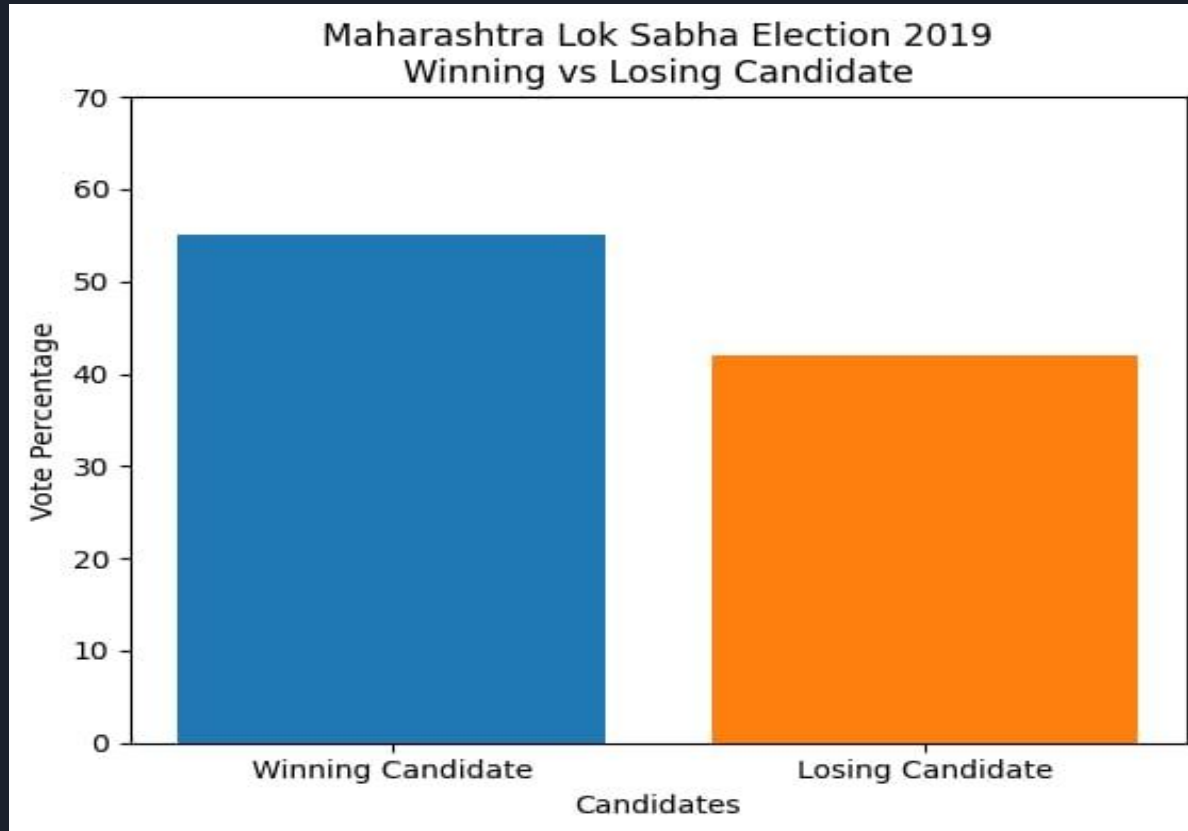
- Uttar Pradesh reflects BJP dominance combined with strong regional party influence.

5. Party-wise Vote Percentage – Pie Chart (Karnataka)

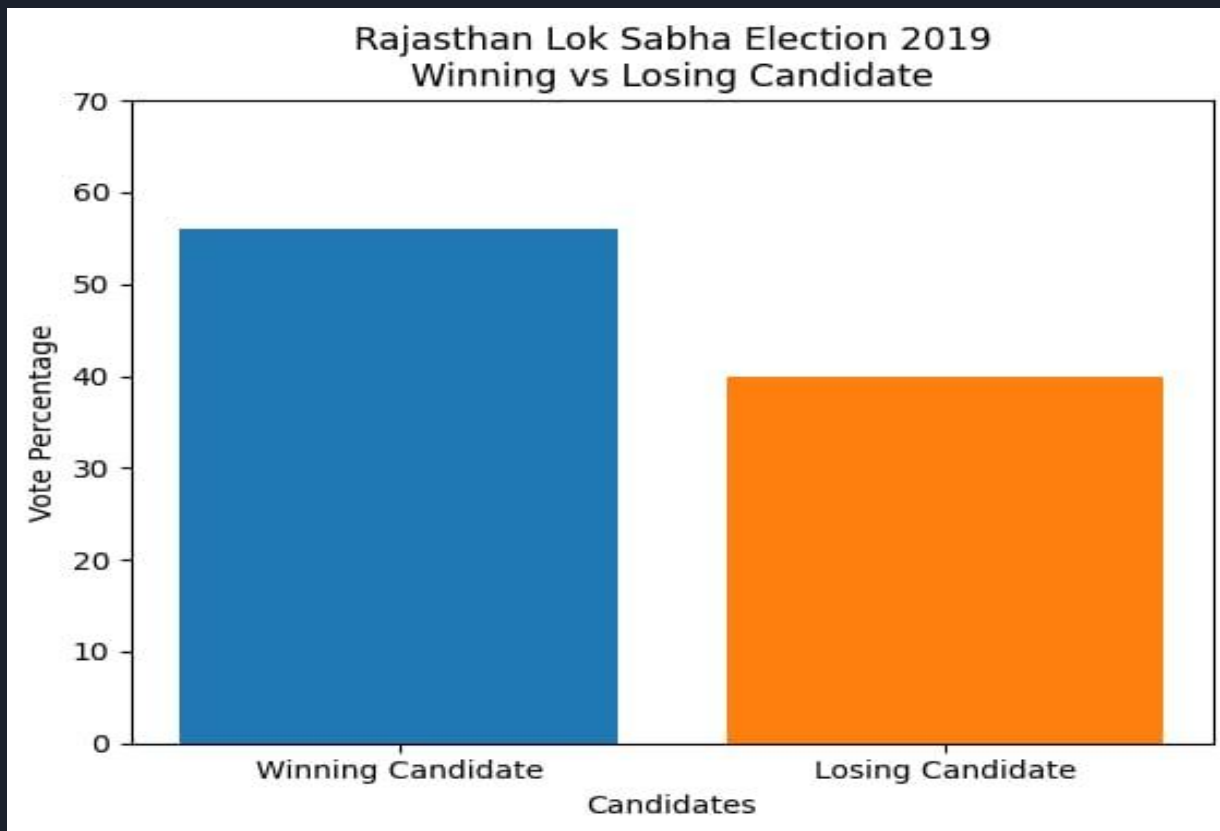


- Karnataka demonstrates a competitive two-party structure, with BJP holding a slight advantage.

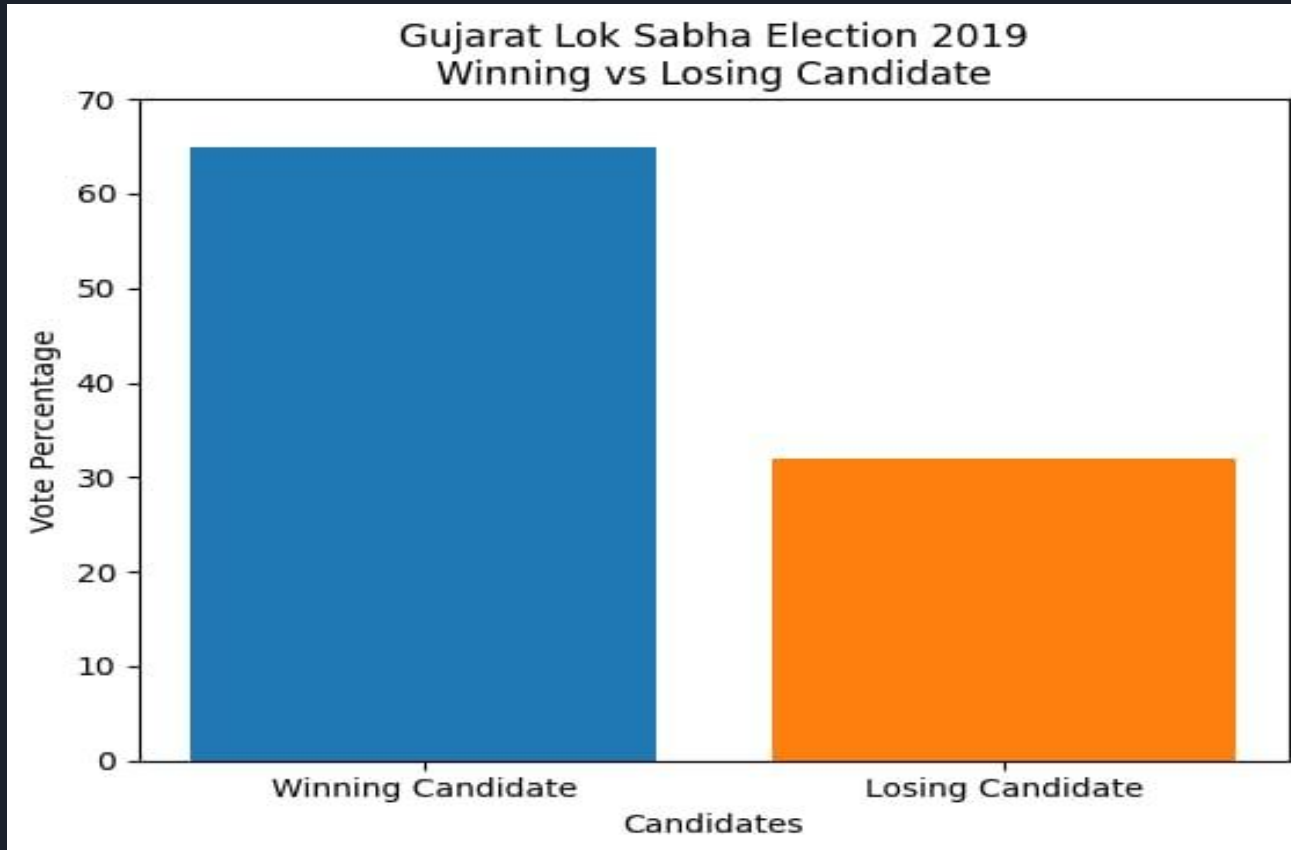
1. Winning vs Losing Candidate – Bar Graph (Maharashtra)



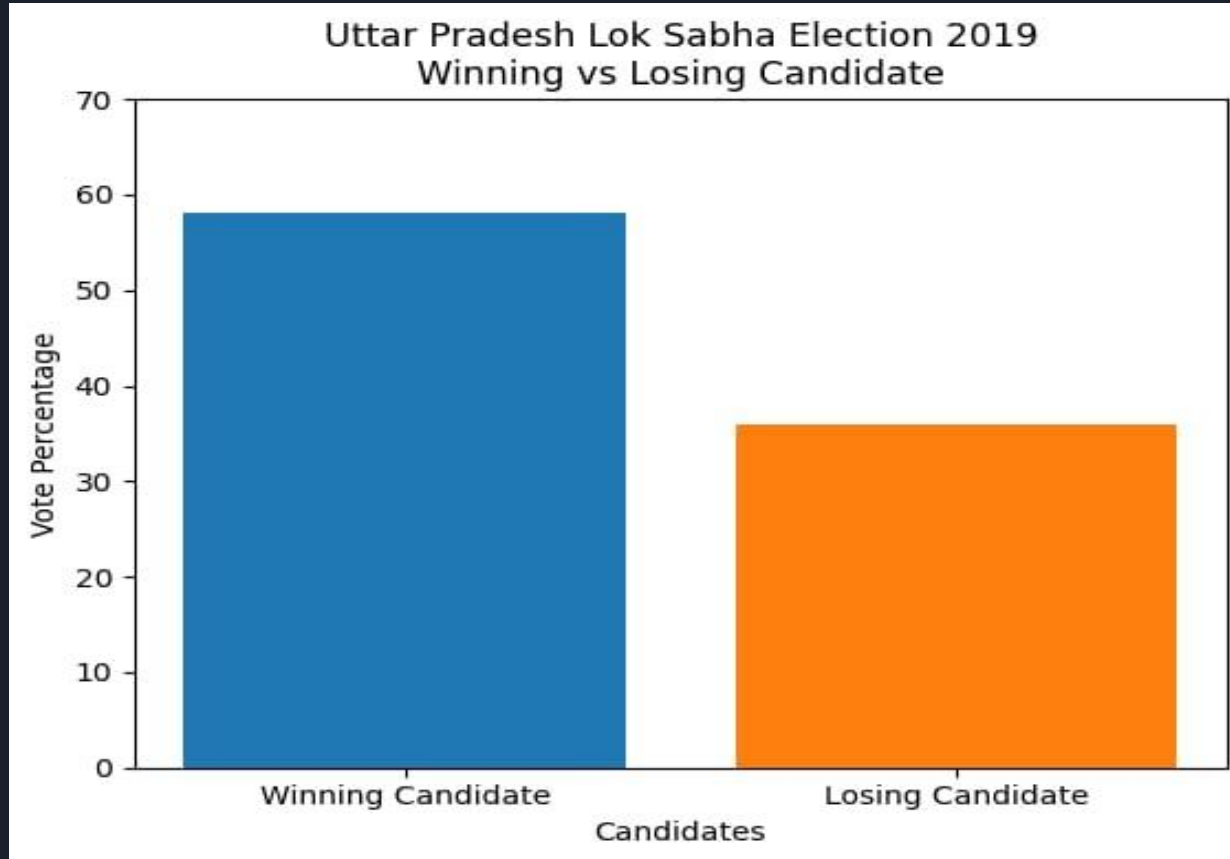
2. Winning vs Losing Candidate – Bar Graph (Rajasthan)



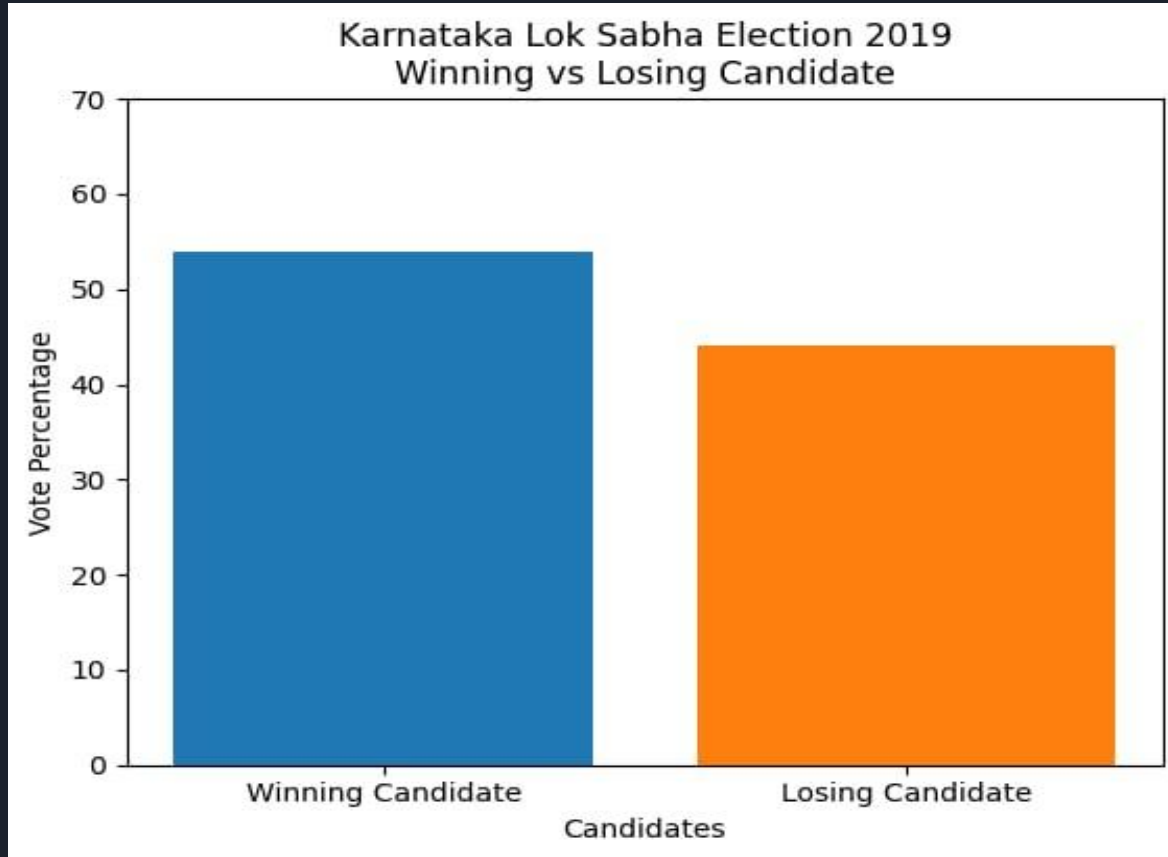
3. Winning vs Losing Candidate – Bar Graph (Gujarat)



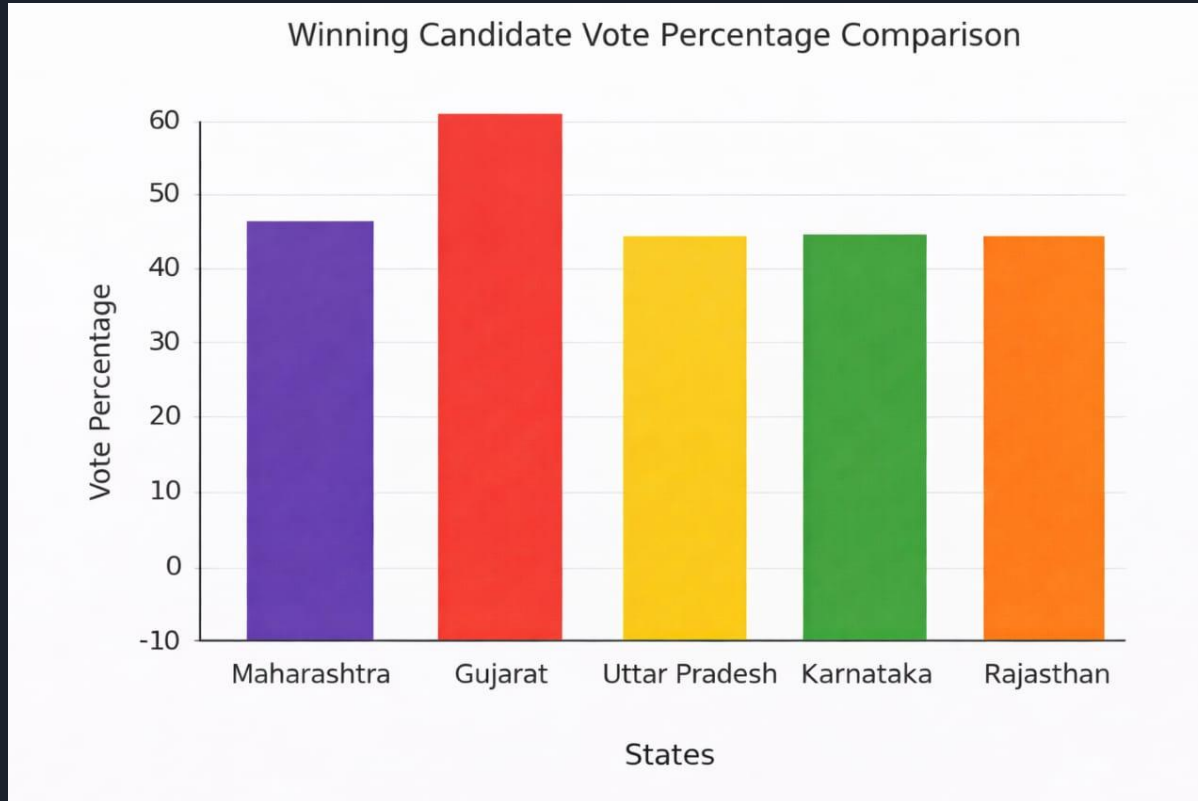
4. Winning vs Losing Candidate – Bar Graph (Uttar Pradesh)



5. Winning vs Losing Candidate – Bar Graph (Karnataka)



Comparison of Winning Candidates Across All Five States (Column Graph)

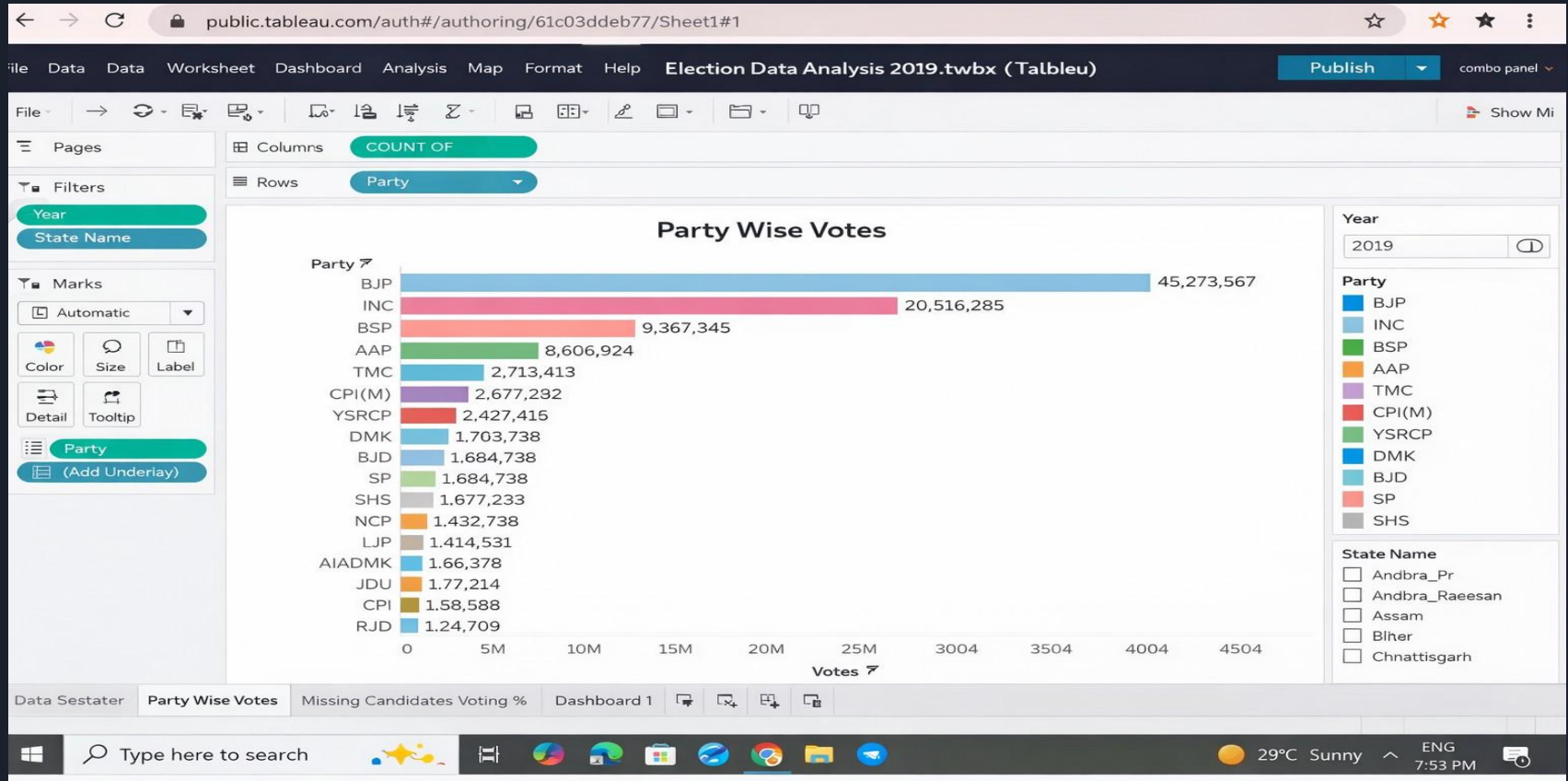


PART 2 – Tableau Visualizations

In this project, Tableau was used to analyze the 2019 election data. Three separate visualizations were created to focus on party-wise vote distribution, comparison between winning and runner-up candidates, and vote percentage of winners. Each visualization provided clear and meaningful insights independently. These visualizations were then combined to create an interactive dashboard. The dashboard allows users to explore election results by state, party, or candidate. Filters and interactive options make data exploration easy and effective. Overall, this approach provides a consolidated, clear, and informative view of party performance, candidate competitiveness, and voting patterns.

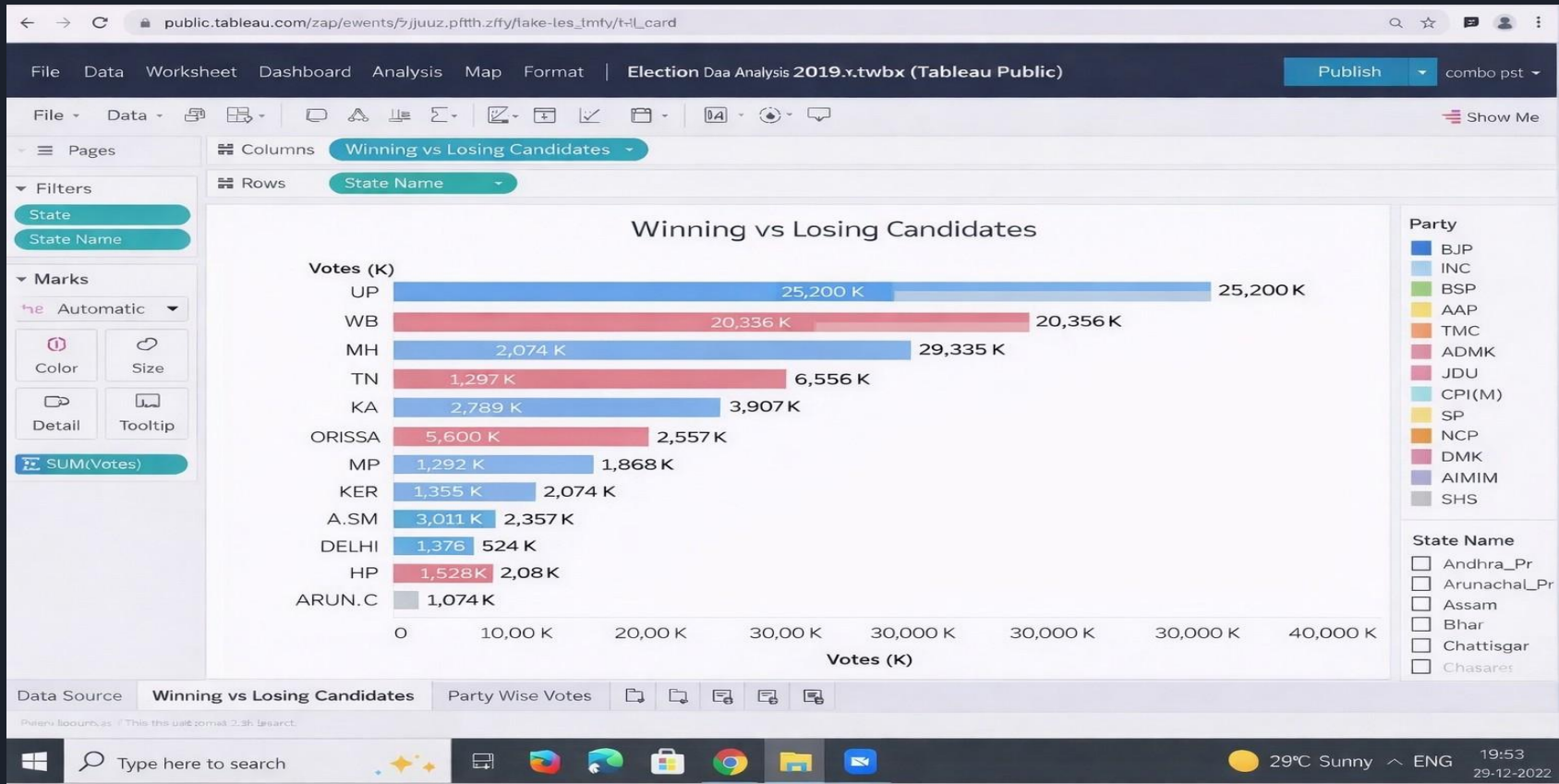
Visualization 1 - Party-wise Votes:

Displays the total votes secured by each political party across the chosen states.



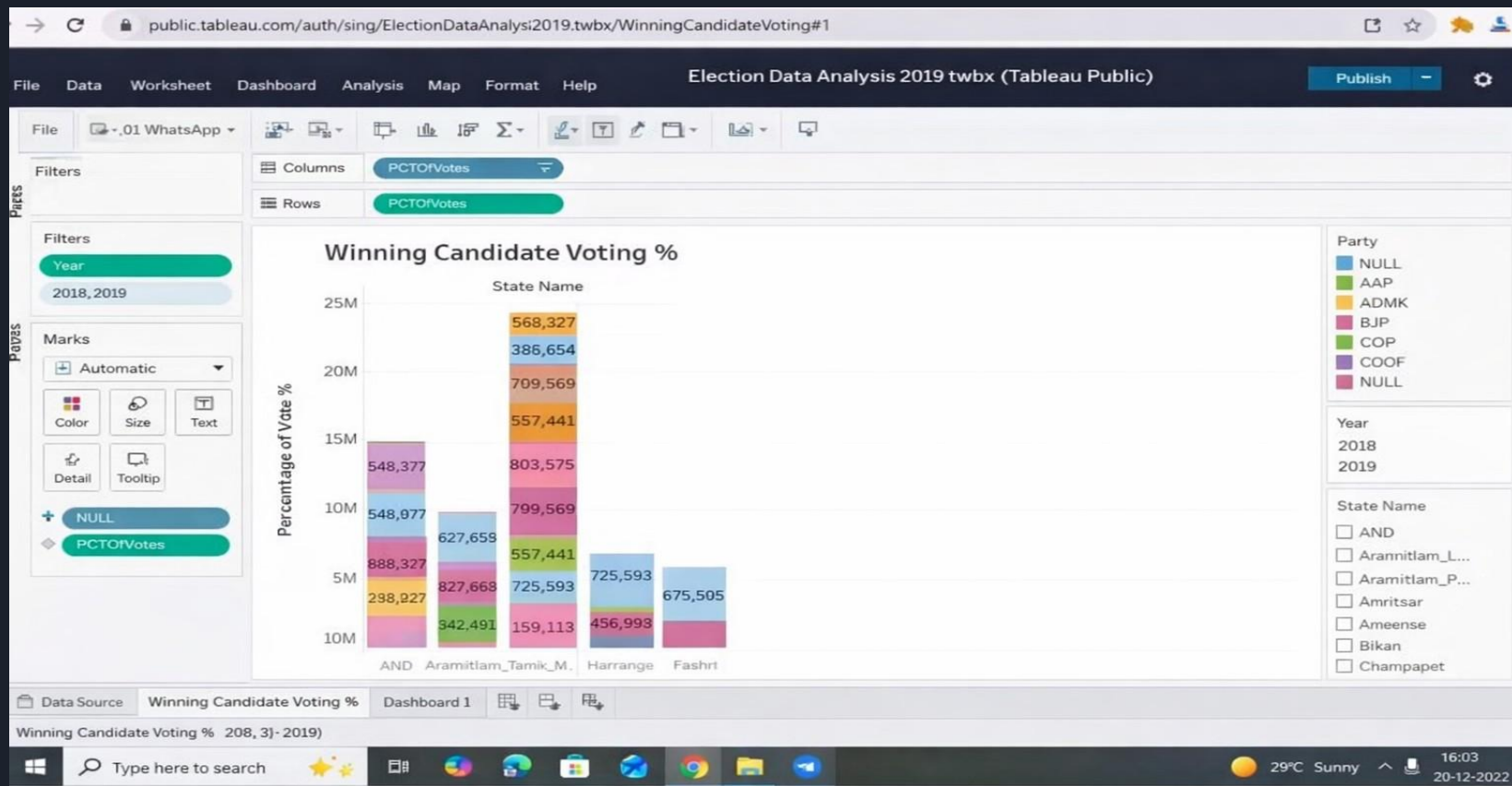
Visualization 2 - Winner vs Runner-Up:

Compares votes of top two candidates in each state to measure competition.

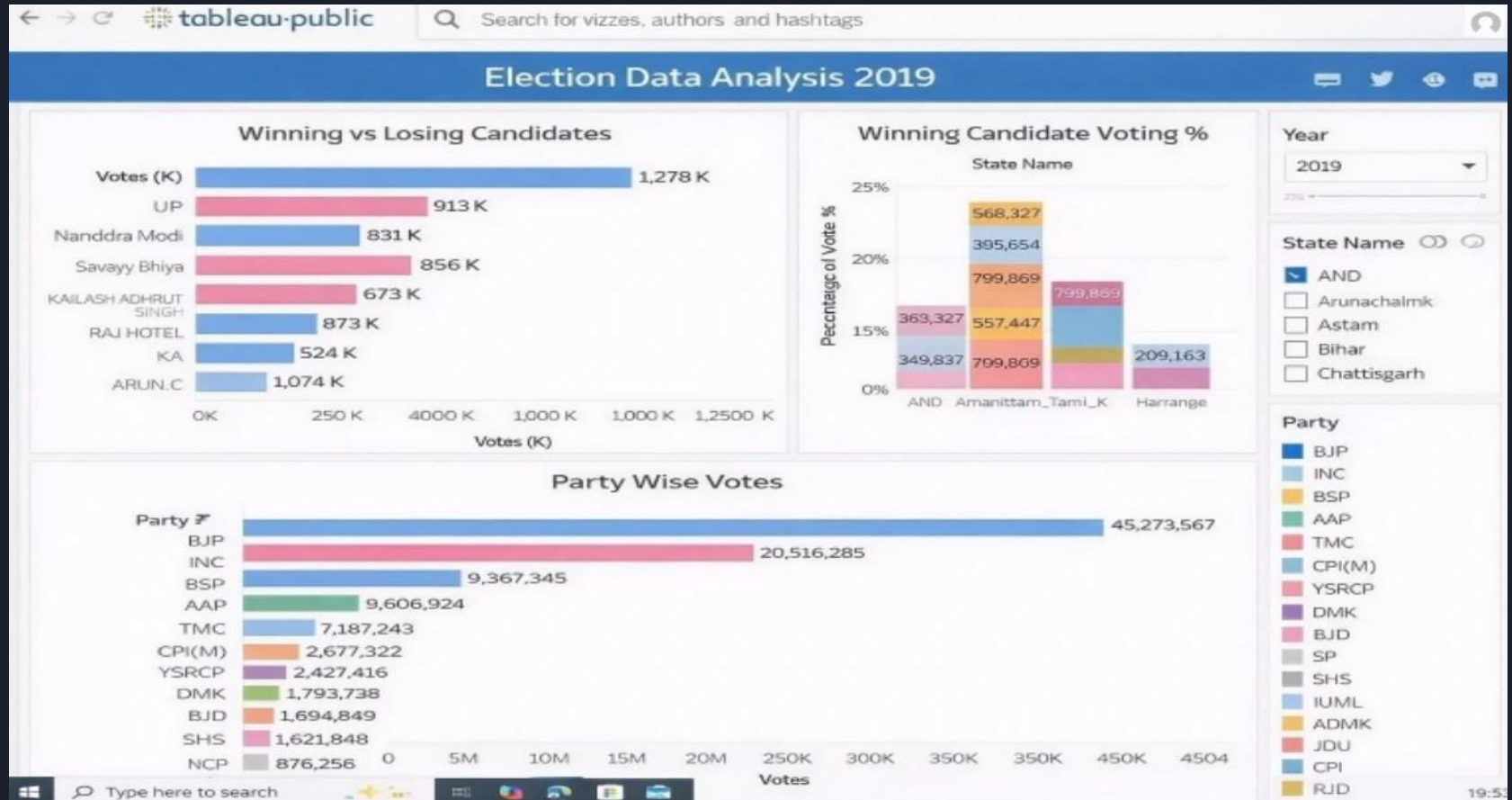


Visualization 3 - Winning Candidates Voting Percentage:

Shows the vote percentage of winners to highlight victory margins



Interactive Data Dashboard using Tableau



PART 3 – BigQuery

- **SQL Analysis Using Google BigQuery – JOIN Query for Year 2019**

```
SELECT
  a.State,
  a.Candidate_Name,
  a.Sex,
  p.Party_Name,
  a.Votes,
  a.Margin_Percentage
FROM
  `your_project.your_dataset.all_states_ge` a
JOIN
  `your_project.your_dataset.ge_parties` p
ON
  a.Party_ID = p.Party_ID
WHERE
  a.Year = 2019;
```

Conclusion

This assignment successfully achieved its objective of understanding election performance using Statistical Charts, Tableau Visualisations and BigQuery SQL Analysis. The analysis clearly presented party-wise distribution, candidate performance comparison, and winning vote strength across Maharashtra, Rajasthan, Kamataka, Gujarat and Uttar Pradesh. Overall, this project enhanced skills in data handling, visualization, interpretation, teamwork, and presentation.

Thank You